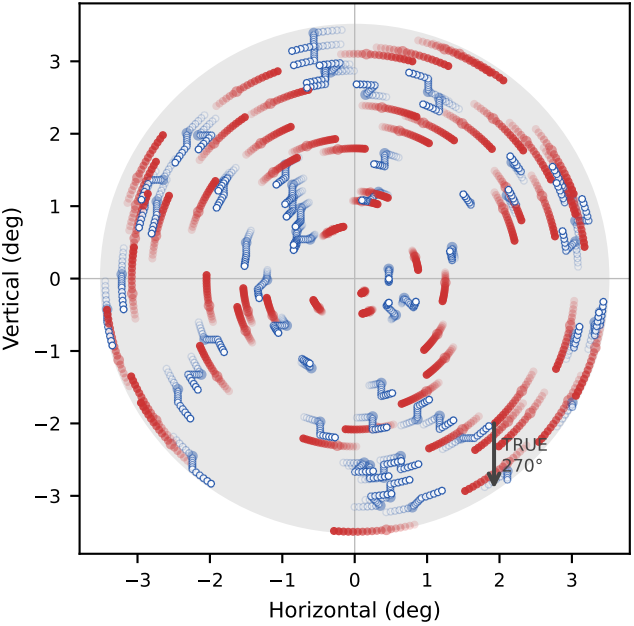
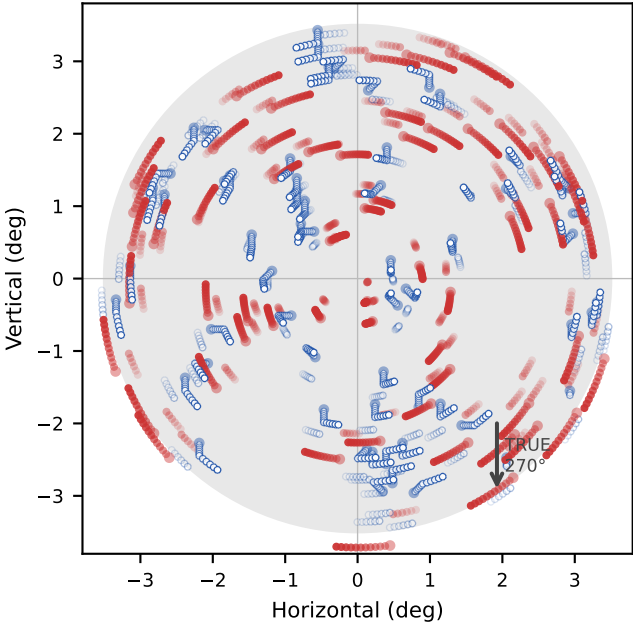


**Stereo-projected trace demo — Z condition, CUED, heading 270° (DOWNWARD)**  
**Comparing: 2D local traces (old method) vs screen-space projections (pre-fix and post-fix)**

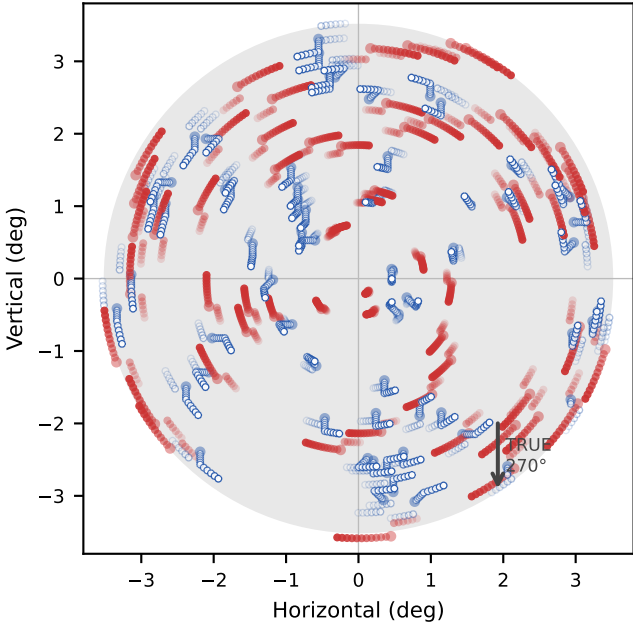
A. 2D local-plane traces  
(old method — no depth projection)  
Artifact: INVISIBLE



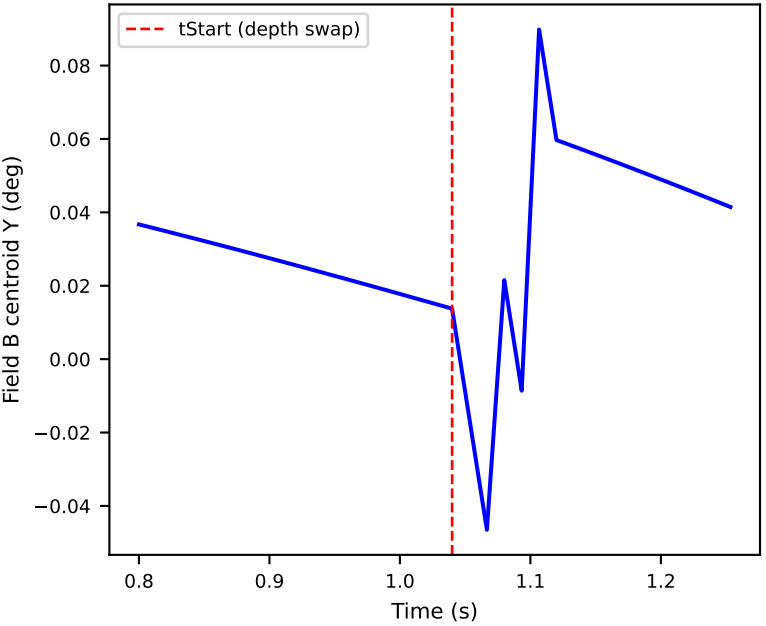
B. Screen-space — PRE-FIX  
(transform.forward pitched 5.0° up)  
Artifact: VISIBLE — upward jump at tStart



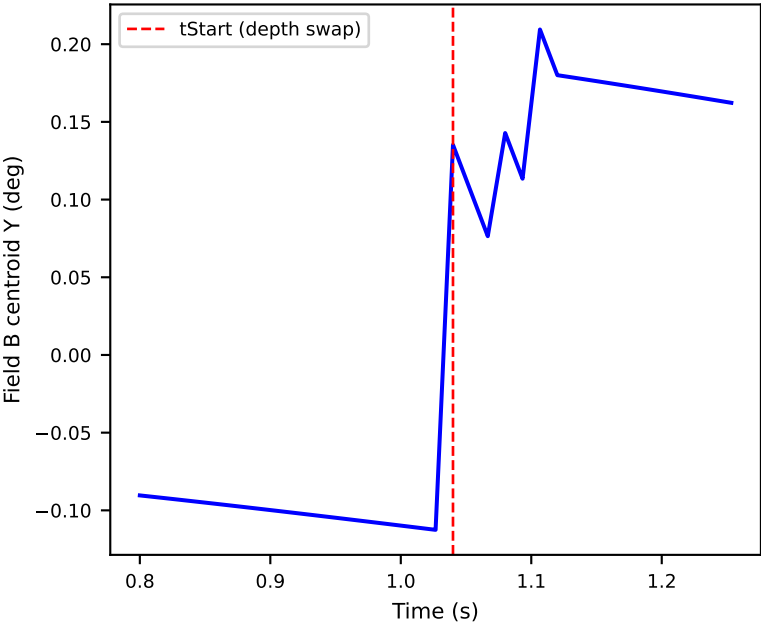
C. Screen-space — POST-FIX  
(camera.forward aligned with gaze)  
Artifact: ABSENT — tiny radial expansion only



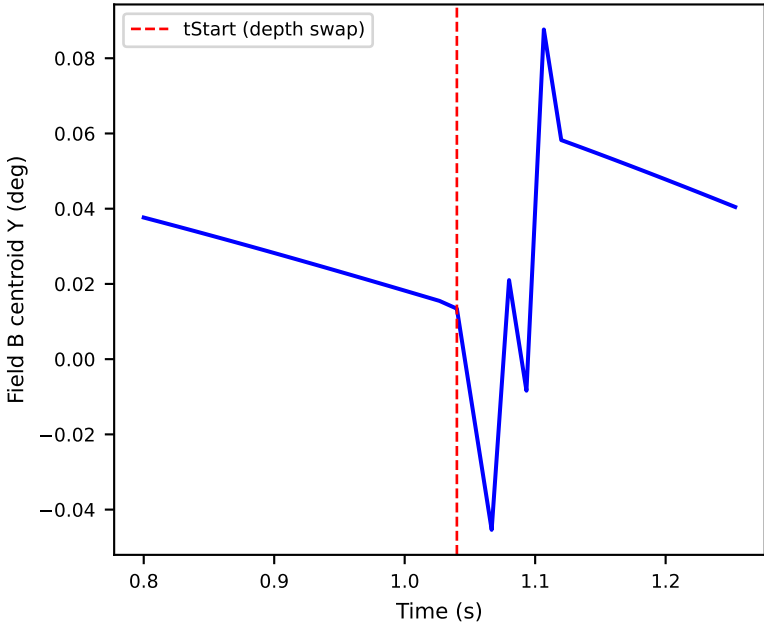
A2. Field B centroid Y — 2D local  
(depth swap invisible)



B2. Field B centroid Y — pre-fix  
(upward jump 5.0° pitch → jump at tStart)



C2. Field B centroid Y — post-fix  
(no directional jump)



● Field A (non-delayed, CW) ○ Field B (delayed, CCW) --- tStart (depth swap)

# Stereo-projected traces: the math and the fix

## What the 2D traces show

The existing trace figures plot dot (x, y) positions in the StimulusBuilder's local plane, in units of metres or degrees of visual angle. Depth is encoded symbolically (filled vs open circles). This is equivalent to an orthographic projection from infinitely far away — it has no perspective and no depth. The depth-offset code (ApplyDepthOffsets) is never called. Result: the figure represents the stimulus \*design\* correctly but is blind to any rendering artifact that lives in the depth-offset code.

## The projection math

For a dot at local position (x, y) [metres] and depth offset z [metres, positive = farther]:

$z\_world = viewDistance + z\_depth$  (total viewing distance)

$x\_screen = x / z\_world$  [radians]    $y\_screen = y / z\_world$  [radians]

binocular disparity =  $IPD / z\_world$  (positive = uncrossed = "far")

Pre-fix artifact (transform.forward pitched  $\theta^\circ$  above gaze):

The forward axis has a spurious upward component:  $\Delta y\_screen = z\_depth \times \sin(\theta) / z\_world$

For  $z\_depth = 0.10\text{m}$ ,  $\theta = 5^\circ$ ,  $z\_world \approx 2\text{m}$ :

$\Delta y \approx 0.10 \times \sin(5^\circ) / 2.0 \approx 0.0044 \text{ rad} \approx 0.25^\circ$

Over 1 frame (13ms at 75 Hz)  $\rightarrow$  apparent velocity  $\approx 0.25^\circ / 0.013\text{s} \approx 19 \text{ deg/sec}$

Translation signal: 2.26 deg/sec. The artifact is  $\sim 8\times$  faster.

## Post-fix behaviour

With camera.forward aligned to the gaze direction, the depth offset has no upward component.

The only residual effect is a tiny perspective-induced radial expansion: dots at the aperture edge ( $2^\circ$  eccentricity, 0.07m) shift outward by  $\approx 0.05^\circ$  when depth changes  $\pm 0.10\text{m}$  at 2m.

This is 2% of the translation signal, radially symmetric, and physically correct.

## How to use these traces as a verification tool

1. For any new design with depth swaps, generate stereo traces before collecting data.
2. Check that the depth-swap frame (tStart) produces ONLY small radial expansion, no directional jump in the centroid time series.
3. If a directional jump is visible, check that Camera.main is correctly set in the scene and that StimulusBuilder has no unexpected rotation in its parent hierarchy.
4. After any engine/build changes, re-run the verification — Unity XR updates can change camera rig hierarchy and tag assignments.